

Task 12: Testing APIs Using Postman

Objective

To test the Node.js + Express REST API using Postman by sending GET and POST requests, validating JSON data, checking HTTP status codes, and verifying error handling.

Tools Used

Node.js, Express.js, Postman

API Endpoints Tested

GET /users

URL: `http://localhost:5000/users`

Expected Status: 200 OK

Purpose: Retrieve all registered users.

POST /users

URL: `http://localhost:5000/users`

Expected Status: 201 Created

JSON Body:

```
{ "name": "Ganga", "email": "ganga@gmail.com" }
```

Error Handling

Sending an empty JSON object `{}` returns HTTP 400 Bad Request with the message 'Name and Email are required'.

HTTP Status Codes

200 OK - Successful GET request.

201 Created - User registered successfully.

400 Bad Request - Missing required fields.

Testing Procedure

1. Started the server using `npm start`.
2. Sent GET request.
3. Sent POST request with valid JSON.
4. Verified response.
5. Sent invalid request to validate error handling.

Screenshots

Insert the following screenshots before submitting:

- Server running terminal
- GET request and response
- POST request and response
- Error handling response

Conclusion

The REST API was successfully tested using Postman. All endpoints returned the expected responses and status codes, confirming correct functionality and reliable error handling.